Sonic Insights Hybrid API

3.0.0 OAS 3.1

/api/v1/openapi.json

A hybrid music intelligence API that combines Spotify user behaviour with a public discovery catalog. It computes a listening fingerprint, detects recent taste drift, supports explainable recommendations, and critiques its own generated insights.

Authorize

Auth

User registration, login, and profile access.

POST	/api/v1/auth/register	Register a new user account
POST	/api/v1/auth/login	Obtain access + refresh token pair
POST	/api/v1/auth/refresh	Exchange a refresh token for a new access + refresh token pair
POST	/api/v1/auth/logout	Invalidate the current access token
GET	/api/v1/auth/me	Get the current user's profile

Ingestion

Import Spotify listening context and the public discovery catalog.

POST	/api/v1/imports/spotify	Start a Spotify import job
GET	/api/v1/imports/jobs/{job_id}	Check import job status
GET	/api/v1/imports/jobs	List your recent import jobs
POST	/api/v1/imports/catalog	Import the public discovery catalog dataset

Listening Events

Record and manage your listening history.

POST `/api/v1/listening-events` Record a new listening event

GET `/api/v1/listening-events` List your listening events (paginated, filterable)

GET `/api/v1/listening-events/stream`
Live stream of listening events via Server-Sent Events

GET `/api/v1/listening-events/{event_id}` Get a single listening event

PATCH `/api/v1/listening-events/{event_id}` Update a listening event

DELETE `/api/v1/listening-events/{event_id}` Delete a listening event

Catalog

Browse and search the imported discovery catalog.

GET `/api/v1/catalog` Search and filter the discovery catalog

GET `/api/v1/catalog/mood-map`
Classify all catalog tracks into mood quadrants with deep statistics

GET `/api/v1/catalog/audio-dna`
Full statistical audio feature distribution across the entire catalog

GET `/api/v1/catalog/genres` Genre breakdown of the imported catalog

POST `/api/v1/catalog/recommend-by-mood`
Natural language mood → ranked catalog recommendations

GET `/api/v1/catalog/{track_id}` Get a single catalog track by ID

GET `/api/v1/catalog/{track_id}/similar`
Find catalog tracks similar to a given track using cosine similarity

Feedback

Full CRUD over user feedback on catalog tracks.

POST `/api/v1/feedback` Create feedback for a catalog track

GET `/api/v1/feedback` List your feedback records

PATCH `/api/v1/feedback/{feedback_id}` Update a feedback record

DELETE `/api/v1/feedback/{feedback_id}` Delete a feedback record

Analytics

Focused hybrid analytics including overview, fingerprint, highlights, and recent change detection.

GET `/api/v1/analytics/overview` High-level hybrid listening summary

GET `/api/v1/analytics/fingerprint` Build your listening fingerprint

GET `/api/v1/analytics/changes/recent` Explain recent taste drift

GET `/api/v1/analytics/highlights` Compact analytics highlights

AI

Explainable hybrid recommendations, grounded insight generation, and critique.

POST `/api/v1/ai/insights` Generate and store a hybrid listening insight

POST `/api/v1/ai/insights/{insight_id}/critique`
Critique a generated insight for grounding and specificity

POST `/api/v1/ai/recommendations/explain`
Return explainable hybrid recommendations grounded in your fingerprint

POST `/api/v1/ai/recommendations/what-if`
Generate counterfactual recommendations for a scenario

MCP

Model Context Protocol server — exposes Sonic Insights tools for AI client integration (Claude Desktop, Cursor, etc.).

GET `/api/v1/mcp/manifest` List all available MCP tools and their input schemas

POST `/api/v1/mcp/invoke` Invoke a named MCP tool with arguments

System

Operational health endpoints.

GET `/health` API health check

GET `/health/detailed` Detailed system health with database statistics

Schemas

AudioDNAFeature > Expand all `object`

Body_login_api_v1_auth_login_post > Expand all `object`

CatalogAudioDNAResult > Expand all `object`

CatalogImportRequest > Expand all `object`

CatalogImportResult > Expand all `object`

CatalogMoodMapResult > Expand all `object`

CatalogSearchResult > Expand all `object`

CatalogTrackRead > Expand all `object`

ChangeMetric > Expand all `object`

EventCreate > Expand all `object`

EventList > Expand all `object`

EventRead > Expand all `object`

EventUpdate > Expand all `object`

FeedbackCreate > Expand all `object`

FeedbackListItem > Expand all `object`

FeedbackRead > Expand all `object`

FeedbackUpdate > Expand all `object`

FingerprintResult > Expand all `object`

FingerprintTraits > Expand all `object`

HTTPValidationError > Expand all `object`

HighlightResult > Expand all `object`

ImportJobRead > Expand all `object`

ImportStartRequest > Expand all `object`

InsightCritiqueIssue > Expand all `object`

InsightCritiqueResult > Expand all `object`

InsightRead > Expand all `object`

MCPInvokeRequest > Expand all `object`

MCPInvokeResult > Expand all `object`

MCPManifest > Expand all `object`

MCPTool > Expand all `object`

MCPToolParameter > Expand all `object`

MoodQuadrantStat > Expand all `object`

MoodRecommendItem > Expand all `object`

MoodRecommendRequest > Expand all `object`

MoodRecommendResult > Expand all `object`

OverviewResult > Expand all `object`

RecentChangesResult > Expand all `object`

RecommendationExplainRequest > Expand all `object`

RecommendationExplainResult > Expand all `object`

RecommendationItem > Expand all `object`

RefreshRequest > Expand all `object`

TokenPair > Expand all `object`

UserCreate > Expand all `object`

UserRead > Expand all `object`

ValidationError > Expand all `object`

WhatIfRecommendationRequest > Expand all `object`